



Washington Hospital Quality Explorer

Analyzing CMS Hospital Quality Performance Across Washington

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Microsoft Excel • PivotTables • Dashboard Development

Healthcare Analytics Portfolio Project



Business Problem

- Healthcare leaders need reliable ways to compare hospital quality performance across Washington and identify opportunities for quality improvement.
- CMS publishes extensive hospital quality data, but it can be difficult to interpret and compare across facilities.



Objective

Develop an executive Excel dashboard that transforms CMS hospital quality data into clear, actionable insights for statewide benchmarking and healthcare quality improvement.



Business Question

How does hospital quality performance vary among Washington hospitals, and what opportunities exist to support **healthcare quality improvement**?



Project Approach

- Filtered CMS Hospital Compare data to Washington hospitals
- Explored trends using Excel PivotTables
- Calculated summary metrics and KPIs
- Designed executive dashboards to communicate findings

From Data to Dashboard

 Import CMS Hospital Compare Data



 Filter & Prepare Washington Data



 Analyze Using PivotTables



 Calculate KPIs & Summary Metrics



 Build Executive Dashboards



 Communicate Business Conclusions

Filter & Prepare – Limited the dataset to Washington hospitals and validated records.

Analyze – Used PivotTables to identify patterns in hospital type, CMS Overall Star Ratings, and statewide quality performance.

Calculate – Created summary metrics to support dashboard KPIs.

Visualize – Designed executive dashboards to communicate findings to decision-makers.



Executive Overview

High-level summary of statewide hospital quality performance and CMS Overall Star Ratings.

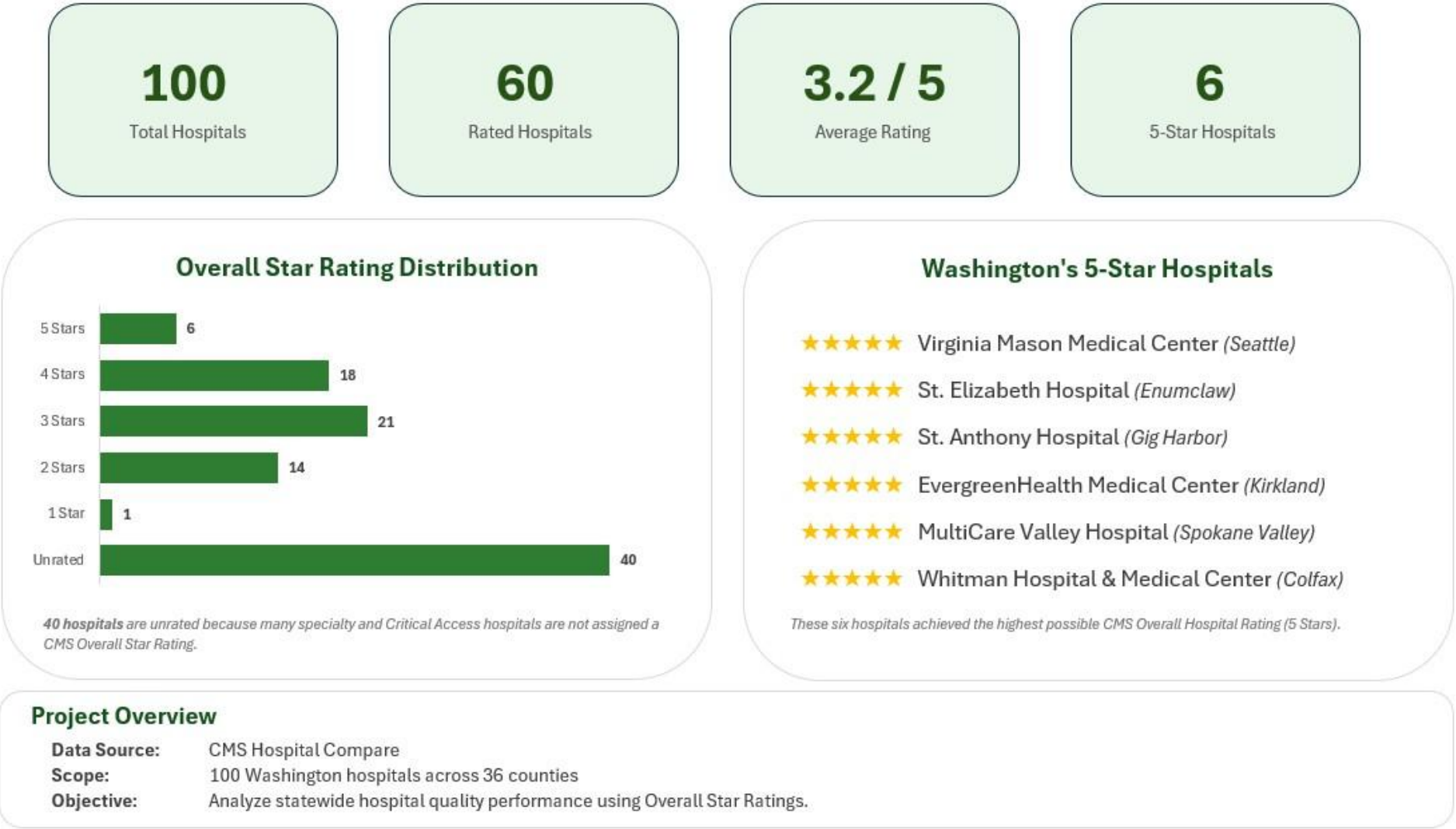


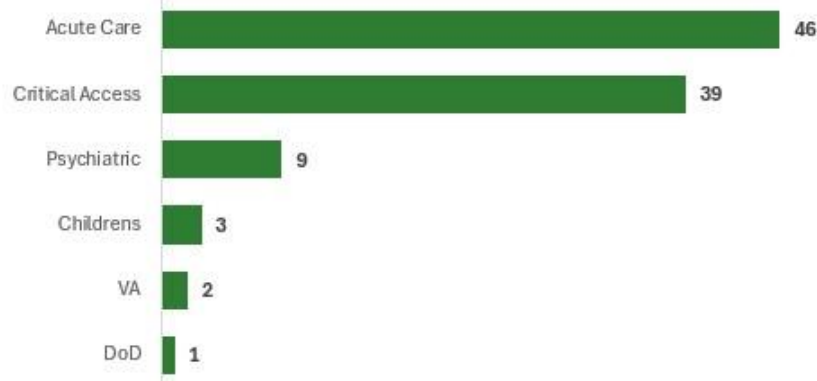
Figure 1. Executive dashboard summarizing statewide hospital quality performance.



Performance Analysis

Supporting analysis highlighting hospital characteristics, statewide quality trends, and business conclusions.

Distribution of Washington Hospital Types



Data Summary

100 Hospitals Analyzed
60 Rated Hospitals
40 Unrated Hospitals
46 Acute Care Hospitals
39 Critical Access Hospitals
6 Five-Star Hospitals
36 Washington Counties Represented

Key Findings

- Acute Care hospitals comprise **46%** of hospitals in the dataset and account for most facilities receiving CMS Overall Star Ratings.
- Critical Access hospitals represent **39%** of Washington hospitals; however, most are **not assigned** an Overall Star Rating.
- Only **six hospitals** achieved the highest possible **CMS 5-Star Rating**.
- **Forty hospitals** were unrated, reflecting CMS eligibility criteria rather than poor performance.
- This dashboard provides a statewide overview of hospital quality performance and highlights variation across facility types.

Business Conclusions

Analysis of 100 Washington hospitals identified meaningful variation in CMS Overall Star Ratings among eligible hospitals. Acute Care hospitals accounted for most rated facilities, while many Critical Access and specialty hospitals were not assigned Overall Star Ratings under CMS methodology. These findings support statewide benchmarking and highlight opportunities to inform healthcare quality improvement efforts through data-driven performance evaluation.

Figure 2. Supporting analysis highlighting hospital characteristics and statewide quality performance.

Key Insights

Analytical Findings

- **Acute Care Hospitals** accounted for nearly half (46%) of all facilities analyzed.
- **Critical Access Hospitals** represented **39%** of hospitals, though many were **not assigned CMS Overall Star Ratings**.
- Only **6 hospitals** achieved the highest possible **5-Star Rating**.
- **40 hospitals** were unrated because they were not eligible under CMS methodology.

Practical Applications

- Executive dashboards simplify interpretation of complex CMS quality data.
- Hospital type should be considered when comparing Overall Star Ratings.
- Statewide benchmarking can help identify performance patterns and inform quality improvement efforts.
- Executive dashboards enable faster, data-informed decision making.

Project Outcome: Developed an executive Excel dashboard that transforms complex public healthcare data into clear, actionable insights.



Project Takeaways

Core Competencies

- Excel Dashboard Design
- PivotTables & PivotCharts
- KPI Development
- Data Visualization
- Healthcare Data Analysis
- Data Storytelling
- Business Communication

Project Reflection

This project strengthened my ability to transform complex public healthcare data into executive-ready dashboards that communicate meaningful business insights. Throughout the project, I balanced analytical accuracy with clear visual communication while ensuring that conclusions remained aligned with the data. The experience reinforced the importance of thoughtful data exploration, stakeholder-focused reporting, and effective data storytelling.